Case 02 · Research and POC: Fire-Sale Cascade Simulator

Following the Magshimim "Research and POC" template: basic research, five key features, and the first POC (goals, plan, design, implementation and conclusions). Every POC was written in Python and actually run, and the numbers here are its output. Each POC states explicitly what is real and what is simulated.

Basic research: sequence of events

- **Define a world:** the user loads or builds a population of agents (funds, banks), each with equity, assets, leverage, and a leverage threshold above which it is forced to act.
- **Shock:** an initial yield rise is injected. The bond price falls according to its duration.
- **Response round:** every agent whose leverage has crossed its threshold sells assets to return to target leverage. An agent whose equity is wiped out sells everything.
- **Price impact:** the round's total sales raise the yield according to market depth, which lowers prices and raises leverage for others.
- **Repeat:** rounds continue until there are no more forced sales, or until an outside actor (a central bank) absorbs the selling.
- **Result:** the yield path, how many agents were forced to sell, how many failed, and by how much the original shock was amplified.

Basic research: system components

- **World Builder:** creates an agent population from data or from distributions.
- **Simulation Core:** the round loop, vectorised (all balance sheets in arrays).
- **Price Impact Model:** the function that turns sales into yield moves.
- **Intervention Module:** a central bank with a purchase cap.
- **Sweep Runner:** runs thousands of random worlds and returns a distribution of outcomes.

Basic research: reliability challenges

- **Parameter sensitivity:** a small change in market depth or leverage thresholds can change the result completely. Ranges must be shown, not a single number.
- **Simplistic behavioural rules:** a real fund can raise capital, get an extension, or sell a different asset. Our model knows only "sell bonds".
- **Calibration:** we do not have the LDI funds' balance sheets from 2022. We can match direction and order of magnitude, not the number.

Basic research: comparison with the leading competitor (Bank of England SWES)

- **Data:** SWES used real, confidential data from the institutions themselves. We work with distributions.

- **Availability:** SWES was a one-off exercise. We are a tool that can be rerun for every "what if".
- **Speed:** an official exercise takes months. A simulation here takes fractions of a second.
- **Reliability:** SWES is far better, because the institutions supplied their own responses.

Conclusion of the comparison: we are not a substitute for an official exercise. We are a fast tool for preliminary exploration. The question is whether a fast tool without real data is useful.

Basic research: ethical issue

A simulation showing "fund X will fail" could, if published about a real institution, itself cause the panic it describes. Response: the tool works on anonymous populations and distributions, and never shows the names of real institutions.

Five key features

- **World building:** Input: number of agents, leverage and threshold distributions, total equity. Output: balance-sheet arrays. How: sampling from distributions with a fixed seed for reproducibility.
- **Round engine:** Input: a world and a shock. Output: yield path, sales, failures. How: a vectorised NumPy loop, until calm or a round cap.
- **Price-impact model:** Input: total sales in a round. Output: a yield move. How: linear in market depth, with a visible parameter.
- **Intervention:** Input: the central bank's purchase cap. Output: absorbed sales and an alternative path. How: netting sales before the price-impact step.
- **Uncertainty sweep:** Input: a range for an unknown parameter (say average leverage). Output: a distribution of outcomes for each value, and the probability of a spiral. How: hundreds of random worlds per value.

The POC: research question

Can a simple agent simulation produce a forced-selling loop that amplifies a shock, as in Britain in 2022? And if we do not know the real exposures, does the tool still give a useful answer?

The POC: goals

- **Reproduce, in direction and order of magnitude, the British episode:** a shock of about 0.6 points that the market amplified to well over a point, and which stopped with an intervention.
- Check that a world without leverage amplifies nothing (a sanity check).
- Run a sweep over an unknown average leverage, and measure whether the outcome changes gradually or jumps.
- Measure performance, to know whether thousands of runs will work in the browser.

The POC: technologies and limits

- Python with NumPy (also available on Pyodide). All agent balance sheets in arrays; a step is a few vector operations.
- **Important limit:** the agents and parameters are synthetic. 60 funds with average leverage of 4, random leverage thresholds between 1.2 and 1.5 times starting leverage, a bond with a duration of 18, and a market depth in which selling 200 units of value moves the yield by one point. The parameters were chosen so the result matches the episode in direction and order of magnitude, not by statistical fit.
- **Out of scope:** multiple asset types, an exposure network between agents, capital raising, interface.

The POC: flow

Build world → initial shock → [compute leverage → forced sales → price impact → if there is a central bank: absorb] → repeat until calm → yield path, sales, failures → sweep across hundreds of worlds.

The POC: results

- **Britain 2022 reproduction without intervention:** a shock of 0.6 points was amplified to 1.13 points within 5 rounds, 1.88 times. 97% of funds were forced to sell.
- **The same world with a central bank absorbing up to 40 units:** the yield stopped at 0.78 points, and 83% of funds still sold. The intervention did not prevent the selling; it broke the loop.
- **Sanity check, a world without leverage:** 0.6 in, 0.6 out. Zero forced sales.
- **The sweep over unknown average leverage (300 worlds per value):** up to average leverage of 4, the chance of the yield jumping above 1.5 points is zero. At 4.5 it is 8%. At 5 it is 80%. At 5.5 it is 100%. In other words, the outcome does not change gradually: it jumps from "safe" to "near-certain spiral" within a range of one point of leverage.
- **Performance:** 1,000 full runs in 0.28 seconds on an ordinary computer. Even if Pyodide is several times slower, a full sweep will run in the browser within seconds.

The POC: what is real and what is simulated

Real: the loop mechanism (leverage, a forced response, selling, price impact) and the threshold behaviour the sweep revealed. Simulated: all agents and parameters, and the Britain reproduction is a fit by eye, not a statistical calibration. "1.88 times" must not be read as a statement about 2022.

The POC: conclusions

✅ Concept proven: a simple model produces the loop, an intervention breaks it, and a world without leverage stays stable. But the sweep revealed something more important: the answer depends almost entirely on one variable we cannot see, the real leverage in the market. Between

4 and 5.5 the tool says both "all fine" and "certain collapse", depending on a value we do not know. The open question for the spec: where will exposure data come from, and what is the tool worth without them?